THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS X — SCIENCE (086) · FULL SYLLABUS PRACTICE PAPER

Session 2026–27 | CBSE / RBSE Pattern | Section A Biology · Section B Chemistry · Section C Physics

Class: 10	Subject: Science	Set 02	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.
2. All sections are compulsory. However, internal choice is provided in some questions. A student is expected to attempt only one of these questions.
3. Draw neat, labelled diagrams wherever required. Use of calculator is not allowed.

SECTION A — BIOLOGY — 16 questions · 30 marks

Q1–Q7 MCQs. Q8–Q9: Assertion (A) / Reason (R) – choose A, B, C or D.

1. Which of the following shows the correct path of air in the human respiratory system? **1**

A. nostrils → pharynx → larynx → trachea → bronchi → bronchioles → alveoli
B. nostrils → larynx → pharynx → trachea → bronchioles → bronchi → alveoli
C. nostrils → pharynx → trachea → larynx → bronchi → alveoli → bronchioles
D. mouth → trachea → pharynx → larynx → bronchi → alveoli → bronchioles

2. After a road accident, Meena can think and speak normally, but she cannot walk in a straight line and cannot pick up a pen accurately. Which part of her brain is most likely damaged? **1**

A. cerebrum B. cerebellum C. medulla D. hypothalamus

3. Which of the following organism–mode of reproduction pairs is correctly matched? **1**

A. Yeast – budding
B. Rhizopus – fragmentation
C. Planaria – spore formation
D. Plasmodium – regeneration

4. A heterozygous tall pea plant (Tt) is crossed with a dwarf pea plant (tt). The percentage of dwarf plants expected in the progeny is: **1**

A. 25% B. 50% C. 75% D. 100%

5. DDT was sprayed on a pond to kill mosquito larvae. After some years its highest concentration is likely to be found in the body of: **1**

A. phytoplankton
B. zooplankton
C. small fish
D. fish-eating birds

6. Which of the following statements about the human heart is correct? **1**

A. The right ventricle pumps oxygenated blood to the body
B. The left atrium receives oxygenated blood from the lungs
C. The left ventricle receives blood from the vena cava
D. The right atrium receives blood from the pulmonary vein

7. The ozone layer is being depleted mainly by the chemicals released from refrigerators and aerosol sprays. These chemicals belong to the group of: **1**

A. hydrocarbons
B. chlorofluorocarbons
C. nitrates
D. sulphates

8. **Assertion (A):** A shoot of a plant bends towards light. **1**
Reason (R): Auxin diffuses towards the shady side of the shoot and stimulates the cells there to grow longer.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

9. **Assertion (A):** Plants have no nervous system but they respond to stimuli. **1**
Reason (R): Plants use chemical (hormonal) means and changes in turgor of cells to respond to stimuli.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

10. State two features of the alveoli that make them suitable for exchange of gases. **2**

11. Differentiate between pollination and fertilisation. **2**

OR

What changes occur in the ovary and in the ovule of a flower after fertilisation?

12. Name two harmful effects of the depletion of the ozone layer on living organisms and the chemical responsible for it. **2**

13. How is food transported in plants? Name the process and the tissue involved, and explain why energy is required for the process. **3**

14. What is a reflex arc? Write two differences between a reflex action and a voluntary action. **3**

15. A pond ecosystem has the food chain: phytoplankton → zooplankton → small fish → large fish → fish-eating birds. The phytoplankton in the pond fix 20 000 J of energy. The pond water was once contaminated with a pesticide, and years later the birds were found to have a very high concentration of the pesticide in their bodies.

(a) Name the producer and one tertiary consumer in the food chain. 1

(b) Why did the birds have the highest concentration of the pesticide? Name this phenomenon. 1

(c) Using the 10% law calculate the energy available to the large fish. 2

OR

(d) What would happen to this pond if all the decomposers were removed? Give two reasons.

16. Explain how urine is formed in the human kidney. Name the organs through which urine is stored and passed out. How does an artificial kidney help patients whose kidneys have failed? **5**

OR

(i) Draw a labelled diagram of a longitudinal section of a bisexual flower. (ii) Explain the process of fertilisation in flowering plants and state what the zygote and the ovary develop into.

SECTION B — CHEMISTRY — 12 questions · 25 marks

Q17–Q21 MCQs. Q22: Assertion (A) / Reason (R) – choose A, B, C or D.

17. A student heats lead nitrate crystals strongly in a dry boiling tube. Reddish-brown fumes are given off and a yellow residue remains. Which of the following correctly describes the change? **1**

A. Thermal decomposition; the fumes are of nitrogen dioxide

B. Displacement reaction; the fumes are of nitrogen

C. Combination reaction; the fumes are of nitrogen dioxide

D. Photolytic decomposition; the fumes are of oxygen

18. An iron nail is kept dipped in copper sulphate solution for about 30 minutes. Which of the following observations and inferences is correct? **1**

A. The blue solution turns pale green because iron is more reactive than copper

B. The blue solution turns colourless because copper is more reactive than iron

C. The green solution turns blue because iron is more reactive than copper

D. No change occurs because iron is less reactive than copper

19. A chemist needs a salt solution that turns phenolphthalein pink. Which of the following salts should she choose? **1**

A. NaCl B. Na_2CO_3 C. NH_4Cl D. KNO_3

20. An oxide X reacts with dilute hydrochloric acid as well as with sodium hydroxide solution, forming a salt and water in each case. X is most likely: **1**

A. Na_2O B. CuO C. Al_2O_3 D. CO_2

21. The molecular formula of the alkene having five carbon atoms (pentene) is: **1**

A. C_5H_{12} B. C_5H_{10} C. C_5H_8 D. C_6H_{12}

22. **Assertion (A):** Aluminium oxide is called an amphoteric oxide. **1**
Reason (R): It reacts with both acids and bases to give salt and water.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

23. Write the balanced equation for the reaction between aqueous sodium sulphate and aqueous barium chloride. State the type of the reaction and the colour of the precipitate. **2**

24. Why should an acid always be added to water, and not water to acid, while preparing a dilute solution? What happens to the concentration of H_3O^+ ions on dilution? **2**

25. Show the formation of magnesium chloride ($MgCl_2$) from magnesium and chlorine using electron-dot structures. Name the type of bond and state its nature. **3**

OR

Give reasons: (a) Sodium is stored under kerosene. (b) Aluminium utensils resist corrosion although aluminium is a reactive metal. (c) Copper does not react with dilute hydrochloric acid.

26. Distinguish between saturated and unsaturated hydrocarbons giving one example of each. Name one chemical test to distinguish between them. **3**

27. In a laboratory, Sneha warms a mixture of ethanol and ethanoic acid with a few drops of concentrated sulphuric acid in a water bath for a few minutes and then pours the mixture into a beaker of water. A sweet, fruity smell is noticed. Later she adds a pinch of sodium carbonate to a test tube of ethanoic acid and observes brisk effervescence.

(a) Name the compound responsible for the sweet smell and the class of compounds it belongs to. 1

(b) Write the chemical equation for its formation. What is the role of concentrated H_2SO_4? 1

(c) Name the gas evolved with sodium carbonate and write the balanced equation of the reaction. 2

OR

(d) Name the reaction that converts an ester back to an alcohol and a sodium salt of the acid by NaOH. State one use of this reaction.

28. Define the following types of chemical reactions and give a balanced equation for each: (a) thermal decomposition, (b) photolytic decomposition, (c) electrolytic decomposition, (d) displacement, (e) double displacement. **5**

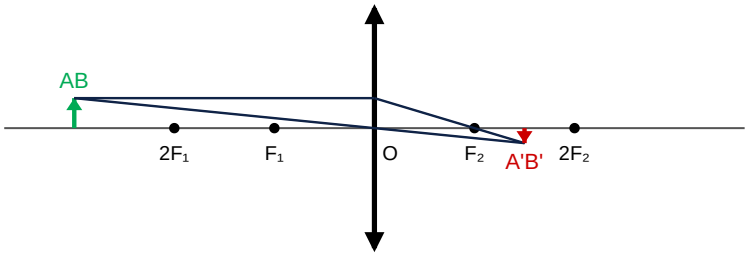
OR

Describe the chlor-alkali process: name the electrolyte, the products formed at each electrode and at the end, give the balanced equation and write one use of each product.

SECTION C — PHYSICS — 11 questions · 25 marks

Q29–Q32 MCQs. Q33: Assertion (A) / Reason (R) – choose A, B, C or D.

29. A candle flame is placed 30 cm from a convex lens of focal length 10 cm (see figure). The image formed is: **1**



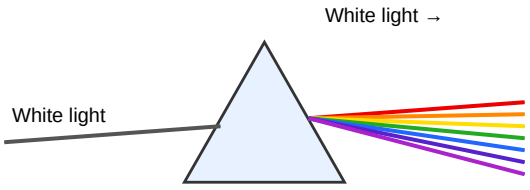
A. real, inverted and diminished, between F_2 and $2F_2$

B. real, inverted and magnified, beyond $2F_2$

C. virtual, erect and magnified

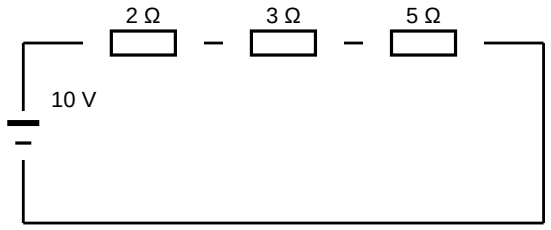
D. real, inverted and of the same size at $2F_2$

30. White light is passed through a glass prism as shown in the figure. The colour of the spectrum that is deviated the most is: **1**



A. red B. yellow C. green D. violet

31. Three resistors of 2 Ω, 3 Ω and 5 Ω are connected in series with a 10 V battery as shown. The potential difference across the 5 Ω resistor is: **1**



A. 1 V B. 2.5 V C. 5 V D. 10 V

32. In India, the AC supply has a frequency of 50 Hz. The direction of the current reverses after every: **1**

A. 1/50 s B. 1/100 s C. 1/25 s D. 1/200 s

33. **Assertion (A):** A fuse wire is made of an alloy having a low melting point. **1**
Reason (R): The heating effect of excess current melts the fuse wire and breaks the circuit.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

34. Two thin lenses of power +3.5 D and −2.5 D are placed in contact with each other. Find the power and the focal length of the combination. **2**

OR

The refractive index of glass with respect to air is 1.5. Define the term refractive index and find the speed of light in glass (speed of light in air = 3×10^8 m/s).

35. A person with hypermetropia can see clearly objects at 50 cm but cannot read at the normal near point of 25 cm. Find the power and nature of the lens required for her to read at 25 cm. **3**

36. An electric bulb is rated 55 W, 220 V. Calculate (i) the current drawn, (ii) its resistance and (iii) the energy consumed in kWh when it is used for 4 hours. **3**

37. A straight wire carries current from the south to the north. (a) State the rule used to determine the direction of the magnetic field around it. (b) How does the strength of the field change as we move away from the wire? (c) What happens to the direction of the field if the current is reversed? **3**

38. Priya uses a convex lens of power +5 D as a magnifying glass to look at a small insect on a leaf. She holds the lens such that the insect is 10 cm from the lens and sees a large upright image of the insect.

(a) Calculate the focal length of the lens. 1

(b) Why does she see a large upright image? Name the nature of the image. 1

(c) Find the position of the image and the magnification produced. 2

OR

(d) At what distance from the lens should the insect be placed to obtain a real image of the same size? Justify your answer.

39. An object 2 cm high is placed 15 cm in front of a concave mirror of focal length 10 cm. Find the position, nature and size of the image. Draw the ray diagram. State one use of a concave mirror. **5**

OR

(a) What is meant by electric short-circuiting and overloading? (b) Name the colour of the live, neutral and earth wires. (c) State the functions of the earth wire and of the fuse in a domestic circuit.